GAS CHROMATOGRAPHY

FORELINE TRAP KIT FOR THE CLARUS SQ 8 Mass Spectrometer



Installation Instructions

PerkinElmer®

Foreline Trap Kit for the Clarus SQ 8 Mass Spectrometer

Installation Instructions

Release History

Part Number	Release	Publication Date
09936442	H	July 2017

Any comments about the documentation for this product should be addressed to:

User Assistance
PerkinElmer
710 Bridgeport Avenue
Shelton, CT 06484-4794

Or emailed to: info@perkinelmer.com

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Introduction

This manual describes how to install and perform maintenance on the foreline trap in a Clarus GC/MS system. More specifically, you will also learn how to install the foreline trap on a roughing pump and replace the activated alumina when necessary.

As with all PerkinElmer GC accessories, this kit includes all of the parts and instructions necessary to install this kit. For complete Clarus GC/MS operating instructions, refer to the *Clarus GC/MS User's Tutorial, or Clarus SQ 8 MS Hardware Guide*, or the *TurboMass Software User's Guide* for your specific model. For complete GC operating instructions, refer to the *Clarus GC User's Manual* for your specific model.

Conventions

In the text, we use three terms in the following standard formats:



WARNING

A warning indicates an operation that could cause *personal* injury if precautions are not followed.



AVERTISSEMENT

Un avertissement indique une opération qui pourrait causer *des blessures si* les précautions ne sont pas respectées.

CAUTION	**A caution indicates an operation that could cause instrument damage if precautions are not followed.**
ATTENTION	**Une précaution indique une opération qui pourrait causer des dommages aux instruments si les précautions ne sont pas respectées.**

Note **Notes emphasize *significant information* in a procedure or description.**

Items in the Foreline Trap Kit (P/N E6420021)

Check all items in the kit against the following list. Report damaged or missing items to the carrier or PerkinElmer to receive the proper credit or replacements.

Item	Part Number (P/N)	Quantity
Extension Tube	09920864	1
Centering Rings	E6071201	2
Clamps	E6071203	2
Trap (and tin of activated alumina)	09923493	1
Installation Instructions	09936442	1

Not included, but available from PerkinElmer is a 0.45 kg container of replacement activated alumina (P/N 09923504).

Major Parts in this Kit

The following figure shows the major parts in this kit with their corresponding part numbers.

Note **The Trap (P/N 09923493) comes with a container of activated alumina.**

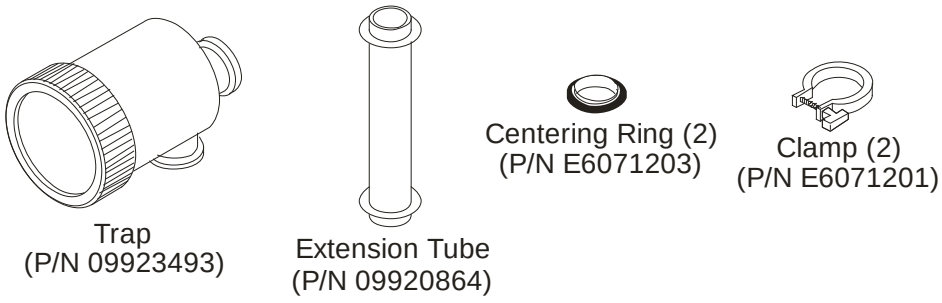


Figure 1. Major parts in this kit.

Installing the Trap

To install the trap read the installation summary below and follow the installation procedures on the pages that follow.

Installation Summary

The following list summarizes the steps necessary to install the trap kit:

Step 1. Prepare the Clarus MS.

Step 2. Install the extension tube.

Step 3. Install the trap.

Step 4. Connect the vacuum line to the trap.

Step 5. Fill the trap with activated alumina.

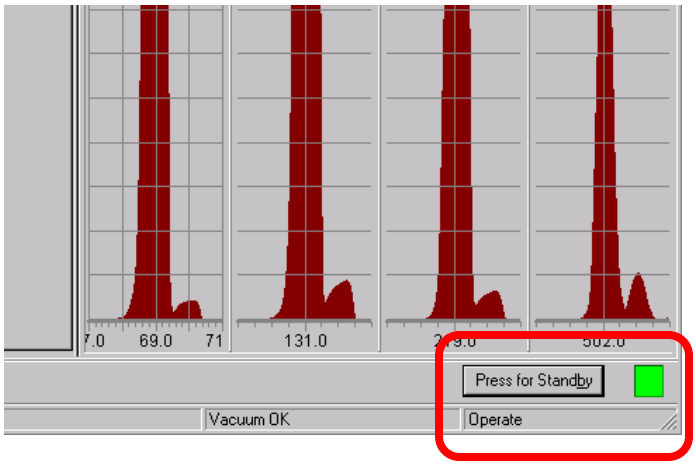
Step 6. Complete the installation.

Step 1. Prepare the Clarus MS

To prepare the Clarus MS for the installation of the trap, follow this procedure:

❶ Turn off the Operate Mode.

1. Display the **Tune** page.
2. If the indicator box next to the **Press for Operate** button is green (indicating an operating instrument), switch it off by clicking on the button.



❷ Cool the Transfer Line and the Ion Source.

CAUTION	**Cool the transfer line and ion source before venting the system. This prevents oxidation of the Clarus MS system components after removing the vacuum**.
ATTENTION	**Refroidir la ligne de transfert et la source d'ions avant d'évacuer le système. Cela empêche l'oxydation des composants du système Clarus MS après avoir retiré le vide.**

1. Select **Heaters** from the **Heaters** menu.

   The Heaters dialogue box appears.

2. Type 20 °C to cool the GC Interface (transfer line) and press **Enter**.
3. On the **Tune** screen, set the Source temperature to 20 °C.
4. Allow the transfer line, column oven, and source to cool before touching them.



WARNING



AVERTISSEMENT

Remember, the transfer line was heated to at least 250 °C and it should take about 10 to 20 minutes to cool.

Rappelez-vous, la ligne de transfert a été chauffée à au moins 250 ° C et cela devrait prendre environ 10 à 20 minutes pour refroidir.

❸ Vent the System.

To vent the turbomolecular pump:

1. Cool the source and inlet to below 100 °C.
2. Select **Vent/Vacuum System Off** from the **Options** menu on the Tune page and confirm that you want to vent the system.
3. Click the **OK** button.
4. If you have a single wide-range gauge, observe the vacuum pressure status on the Tune screen.

   ➢ If you have a two gauge display, observe the Vacuum Pressure Gauges status on the Tune screen.

   - The Penning gauge displays **FAIL** after the turbopump decreases to 50% speed and the vent valve is opened.
   - The Pirani gauge displays decreasing vacuum by first displaying **OFF** and then **FAIL**.

Note **Venting the MS may take several minutes depending on the rate of helium flow into the Clarus MS manifold.**

5. The system is now vented to atmosphere (or optional dry gas).

Step 2. Install the Extension Tube

CAUTION	**Before disconnecting tubing, make sure that the Clarus MS vacuum pumps are NOT running and the Clarus MS is vented to atmosphere**.
ATTENTION	**Avant de débrancher le tube, assurez-vous que les pompes à vide Clarus MS ne fonctionnent PAS et que le Clarus MS est ventilé dans l'atmosphère..**

To install the extension tube on the roughing pump, follow this procedure:

1. Loosen and remove the clamp that holds the vacuum line to the roughing pump connection.

CAUTION	**Ensure that you are connecting the extension tube to the inlet port, as shown in Figure 2. NEVER connect tubing to the exhaust port**.
ATTENTION	**Assurez-vous que vous raccordez le tube d'extension au port d'entrée, comme indiqué sur la Figure 2. NE JAMAIS brancher le tube sur l'orifice d'échappement.**

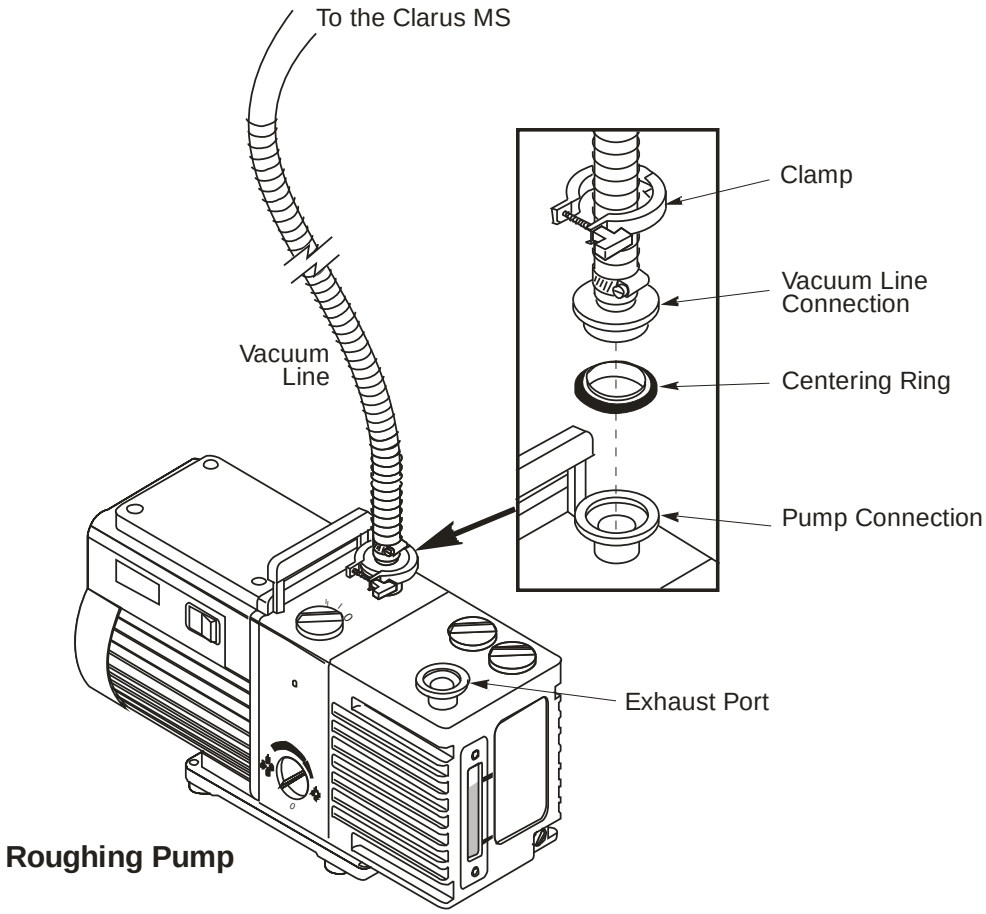


Figure 2. Disconnecting the vacuum line from the roughing pump.

2. Clean the roughing pump inlet connection and centering ring of any particulates.
3. Position the centering ring on the end of the extension tube.
4. Insert a clamp over the extension tube and secure the extension tube to the roughing pump.

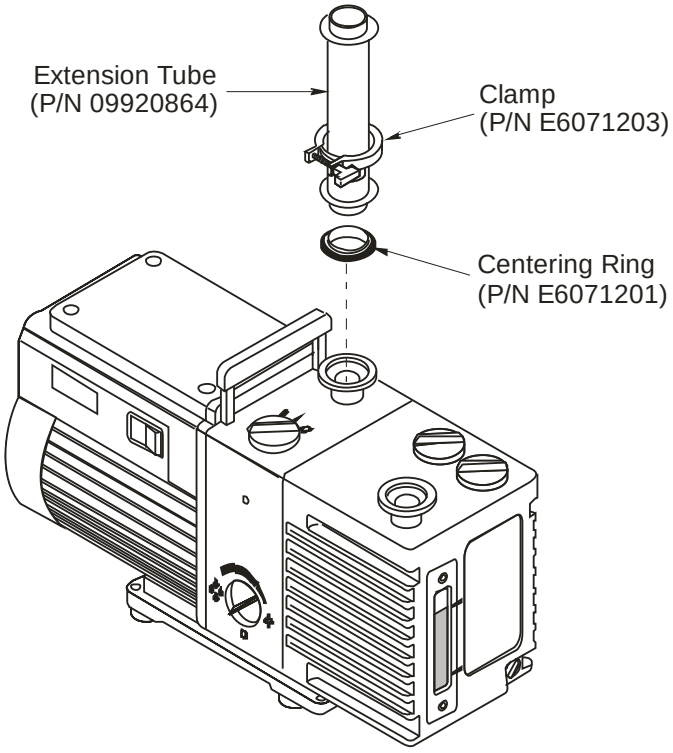


Figure 3. Installing the extension tube on the roughing pump.

Step 3. Install the Trap

To install the trap, follow this procedure:

1. Insert another clamp over the end of the extension tube.
2. Insert a centering ring on the end of the extension tube.
3. Insert the trap on the extension tube.
4. Secure the trap to the extension tube with the clamp.

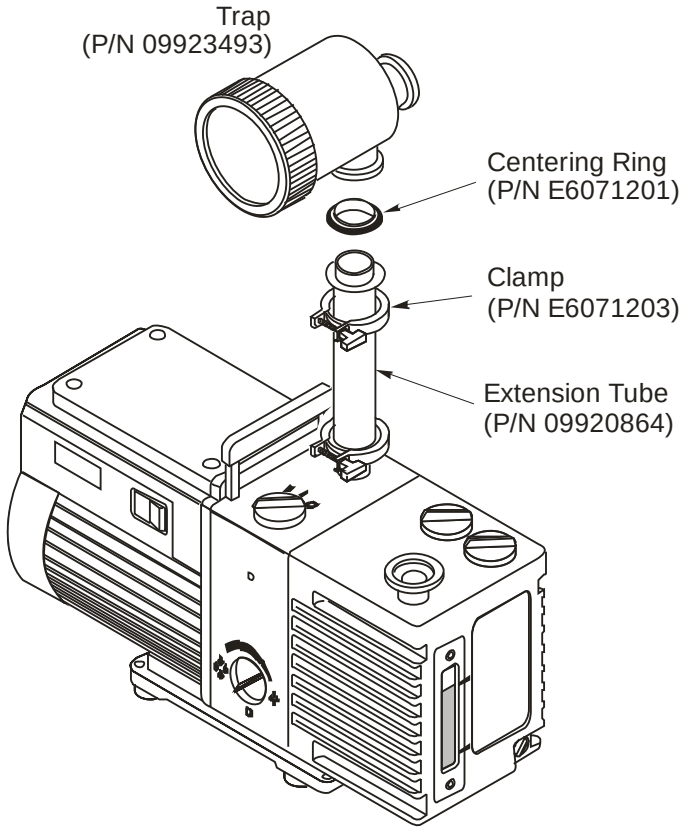


Figure 4. Installing the trap.

Step 4. Connect the Vacuum Line to the Trap

The following two procedures describe how to connect the vacuum line to the trap. Select the procedure that describes your system.

On a New System

To connect a vacuum line to the trap on a new system, follow this procedure:

1. Insert a centering ring on the connecting flange end of the vacuum line.
2. Insert a clamp over the end of the vacuum line.
3. Secure the vacuum line to the trap with the clamp.

On an Existing System

When connecting the vacuum line to the trap on an existing system, the vacuum line may have some oil contamination near the connecting end, which must be removed.

To prepare the vacuum line on an existing system, remove and replace the entire vacuum line (P/N E6400223) to the Clarus MS ***or*** cut off the contaminated part of the vacuum line as follows:

1. Using a flat-blade screwdriver, loosen the clamp that secures the connecting flange to the vacuum line.
2. Remove the connecting flange from the end of the vacuum line.
3. Cut 6 to 12 inches off the vacuum line.
4. Carefully wipe any oil from the flange with a lint-free laboratory wipe (Kimwipe).
5. Reconnect the flange to the vacuum line and secure it by tightening the clamp with a flat-blade screwdriver.

To connect a vacuum line to the trap, follow this procedure:

1. Insert a centering ring on the connecting flange end of the vacuum line.
2. Insert a clamp over the end of the vacuum line.
3. Secure the vacuum line to the trap with the clamp.

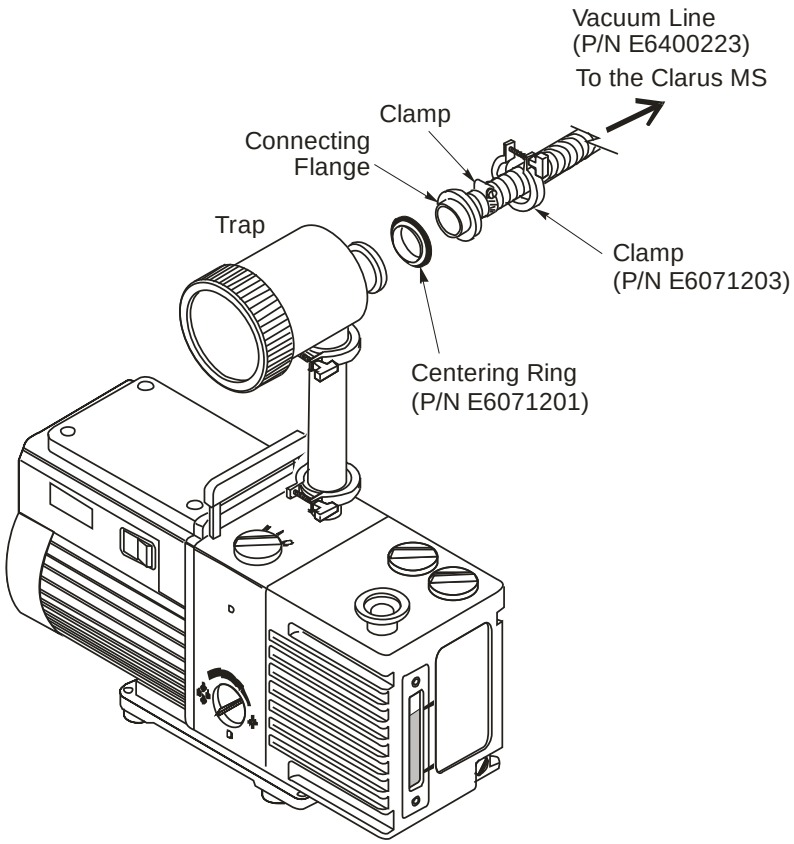


Figure 5. Connect the vacuum line to the trap.

Step 5. Fill the Trap with Activated Alumina

Before using the trap, fill the basket inside the trap with activated alumina. Replacement activated alumina is available from PerkinElmer (P/N 09923504).

To remove the basket from the trap and fill it with activated alumina, follow this procedure:

WARNING	**Prevent inhalation, eye contact, and unnecessary skin contact of alumina dust.** **Use gloves to prevent fingerprint contamination of those areas inside the vacuum system.** **Activated alumina is a desiccant that absorbs materials from your Clarus MS system. If you try to reactivate alumina by baking it, make sure you bake it in a well-ventilated fume hood to protect yourself from the harmful materials released during baking.**

AVERTISSEMENT	**Empêcher l'inhalation, le contact avec les yeux et le contact inutile avec la peau de la poudre d'alumine.** **Utilisez des gants pour éviter la contamination de l'empreinte digitale de ces zones dans le système d'aspiration.** **L'alumine activée est un déshydratant qui absorbe les matériaux de votre système Clarus MS. Si vous essayez de réactiver l'alumine en la cuire, assurez-vous de la cuire dans une hotte bien ventilée pour vous protéger contre les matériaux nocifs libérés lors de la cuisson..**

1. Unscrew the large nut on the trap by turning the nut counterclockwise until it is free from the trap body. The lid should remain inside the nut.

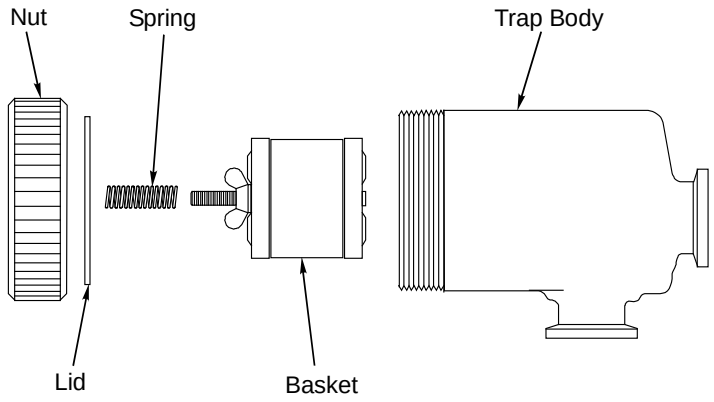


Figure 6. Removing the basket from the trap body.

2. Remove the spring from the threaded rod on the basket.
3. Grasp the threaded rod near the wing nut and pull the basket from the trap body.
4. Place the basket on a clean, flat surface. Then remove the wing nut, washer, end cap, compression spring, and perforated disk.

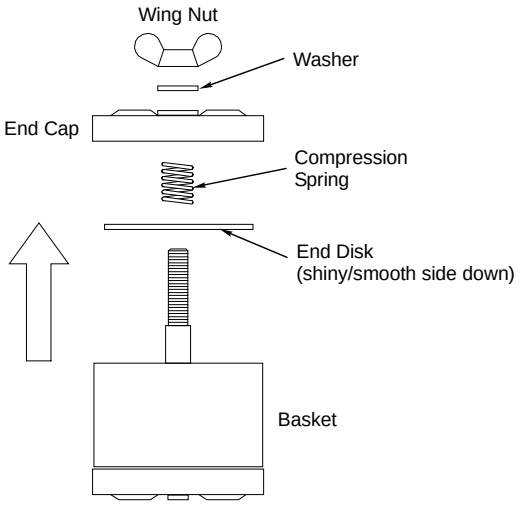


Figure 7. Removing the end cap from the basket.

5. Fill the basket with activated alumina to a level equal to the mark on the spindle. Gently tap the basket to settle the activated alumina. Then add more activated alumina if necessary.

    - If you are performing maintenance to change the alumina, you will need to empty and clean the basket before adding fresh alumina (P/N 09923504).

CAUTION	**Do not overfill the basket with activated alumina. The rotary pump can pull in crushed alumina, thereby causing damage to the pump**.
ATTENTION	**Ne pas trop remplir le panier avec de l'alumine activée. La pompe rotative peut tirer de l'alumine écrasée, causant ainsi des dommages à la pompe.**

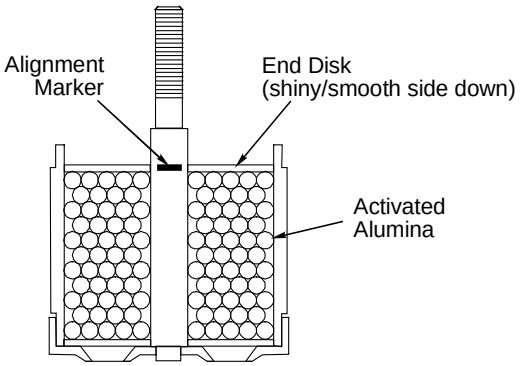


Figure 8. Filling the basket with activated alumina.

6. Position the end disk over the spindle with the shiny/smooth side facing (touching) the activated alumina.
7. Slide the compression spring, end cap, washer, and wing nut over the spindle.

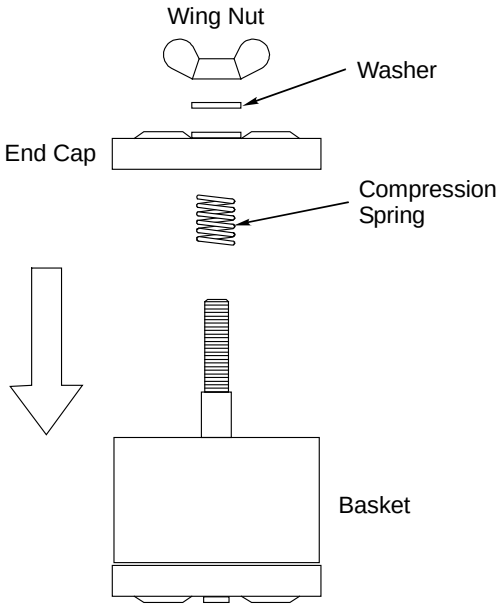


Figure 9. Reassembling the basket.

8. Insert the basket into the trap body.
9. Slide the spring over the spindle, then replace the nut on the trap body by turning it clockwise until it is finger tight.

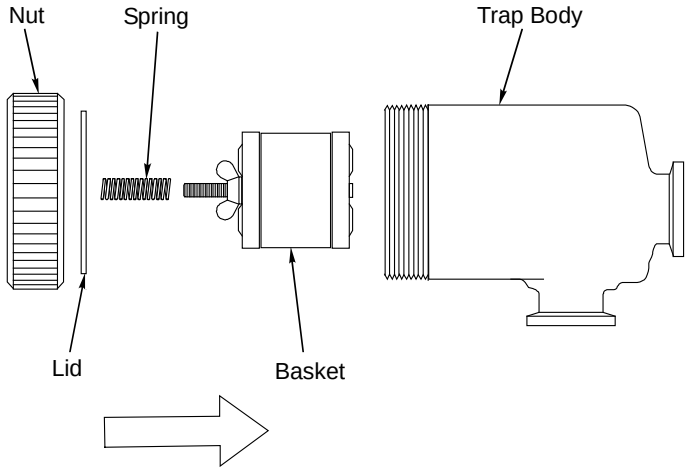


Figure 10. Replacing the basket in the trap.

Step 6. Complete the installation

To complete the installation, follow this procedure:

1. Make sure the vacuum lines are correctly installed.

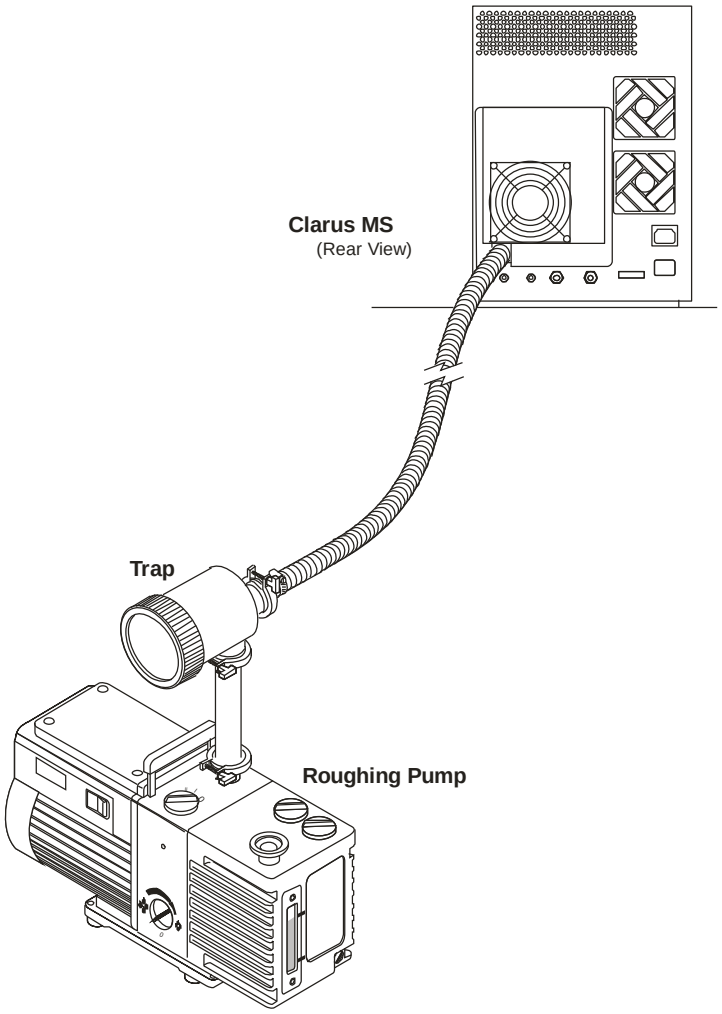


Figure 11. Trap installed on the roughing pump and connected to the rear of the Clarus MS.

2. Turn on the Clarus MS vacuum pumps and check for leaks.
   - For detailed leak-checking instructions, refer to Chapter 4 in the *Clarus MS Hardware Guide*.

Pumping Down a Turbomolecular Pump Vacuum System

➢ Select **Pump/Vacuum System On** from the **Options** menu on the Tune page.

The menu name will change from **Pump/Vacuum System On** to **Vent/Vacuum System Off**, and the system will begin its pump-down sequence. Once **OPERATE** is enabled, it remains enabled unless the **Vent/Vacuum System Off** command is given.

Maintenance

During normal use, the activated alumina will change to a brown color. When the entire basket full of alumina turns brown, it is time to change the alumina. The time period varies depending on your system use, sample type, etc. A typical period is six months. To maintain optimum performance, we recommend changing the activated alumina every three months.

Replacement Alumina

Not included, but available from PerkinElmer is a 0.45 kg container of replacement activated alumina (P/N 09923504).

Changing the Alumina

To change the alumina, follow the procedure in this manual starting with "Step 1. Prepare the Clarus MS" on page 3 of this manual.



WARNING

Do not bake in your GC oven alumina that has been exposed to oil and/or contains oil from the rotary vane pump. This will contaminate your GC oven. Only bake new alumina in your GC oven.



AVERTISSEMENT

Ne pas cuire dans votre alumine de four GC qui a été exposée à de l'huile et / ou contient de l'huile de la pompe à girouette. Cela contaminera votre four GC. Cuire seulement de l'alumine dans votre four GC.

To prevent increased background levels of water, new alumina must be conditioned prior to filling the foreline trap basket. This can be achieved by placing the alumina in a beaker and baking it in your GC oven for 2 hours at 250 °C. After baking, cool the alumina to room temperature before filling the trap.





PerkinElmer
710 Bridgeport Avenue
Shelton, CT 06484-4794, U.S.A.

Internet: http://www.perkinelmer.com
email: info@perkinelmer.com

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